

Hrishikesh Kurapati

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SUMMARY

Computer Science undergraduate with strong foundations in data structures, algorithms, and scalable system design. Experienced in developing end-to-end, production-ready software and AI-powered systems, from backend architecture to deployment. Passionate about building high-impact systems through clean abstractions, performance-focused engineering, and rigorous problem solving.

EDUCATION

Indian Institute of Information Technology , Sri City, Andhra Pradesh	2023 - Present
B.Tech, Computer Science and Engineering	CGPA: 8.73
Resonance Junior College , Telangana	2021 - 2023
Intermediate, 10-12th, Science	CGPA: 9.63

PROJECTS

LeanFeastAI — End-to-End AI Culinary Platform

August 2025 - December 2025

- **Distributed Systems Reliability:** Built a production-ready backend service using **FastAPI** and **PostgreSQL**, adhering to **SOLID principles**; implemented **event-driven orchestration** with **AsyncIO** and **HTTP connection pooling** to manage concurrent AI workloads, reducing user-perceived latency by 40% while ensuring 99.9% reliability via **exponential backoff** retry mechanisms.
- **AI & RAG Architecture:** Architected a high-fidelity **Retrieval-Augmented Generation (RAG)** pipeline using **LangChain** and **Pinecone**, orchestrating **Gemini 2.5** to synthesize personalized meals; developed an **intelligent optimization engine** utilizing **LLM-driven context injection** to dynamically adapt recipes for specific user-defined goals (e.g., Keto, Vegan).
- **Full-Stack Product Engineering:** Engineered a seamless **hands-free culinary interface** in **React** integrating **Web Speech API** and **Porcupine** for custom wake-word detection (“LeanFeast”); built a scalable **Realtime Community Hub** using **Supabase Realtime** and **Redux Toolkit**, implementing **optimistic UI** updates to handle high-frequency user engagement.

CDRP using GNNs — Deep Learning for Precision Oncology

May 2025 - Dec 2025

- Achieved superior predictive performance of 0.9516 AUC (vs 0.9436 baseline) by integrating custom representation modules into GraphCDR, while delivering a 14× training speed-up (reducing runtime from ~50 mins to ~3 mins).
- Developed custom node representation modules using **Graph Isomorphism Networks (GIN)** for molecular graphs and **Cross-Modal Attention** to fuse high-dimensional multi-omics cell-line data.
- Optimized the GCLM-CDR architecture with modified representations, reducing computational overhead and cutting total training time by 36% (saving >1.5 hours per run).
- Benchmarked **4 GNN architectures** (GIN, GCN, GAT, GraphSAGE) and **3 multi-omics fusion strategies** (VAE, FC, Attention) through targeted ablation studies to identify the optimal node representation modules.

CollabSync — End-to-End Influencer Marketing Ecosystem

Jan 2025 - Present

- Engineered a complete campaign management system with complex React forms and multi-step validation.
- Built a full-featured e-commerce marketplace with session-based shopping cart and secure checkout.
- Architected backend aggregation logic using MongoDB/Express for dynamic leaderboards and inventory management.
- Developed data-driven dashboards integrating Chart.js for audience demographics and performance analytics.
- Spearheaded migration from legacy EJS to a modern React SPA architecture, enhancing maintainability.

SKILLS

Languages	C, C++, Java, Python, JavaScript, TypeScript
Systems & HPC	CUDA, OpenMP, MPI, Distributed Computing, Linux/Unix, AWS (SageMaker, S3)
Machine Learning & AI	PyTorch (Geometric), TensorFlow, LangChain, RAG, LLMs, NLP, Computer Vision
Web & Databases	React.js, Node.js, FastAPI, JWT, OAuth, Sessions, SQL, PostgreSQL, Supabase, MongoDB, Pinecone, Redux Toolkit
Algorithms & Data	Data Structures, OOP, Pandas, NumPy, Scikit-learn

CERTIFICATIONS

- [Deep Learning Specialization, DeepLearning.AI \(Coursera\)](#)
- [The Web Developer Bootcamp 2025, Udemy](#)